Heart Disease Prediction: A Machine Learning Approach to Enhancing Healthcare Outcomes

Problem Description:

In the realm of healthcare, heart disease stands as a predominant cause of mortality worldwide. Its early detection is paramount in reducing the risk of severe complications and fatalities. This project aims to harness the power of machine learning to predict heart disease, a critical aspect of proactive cardiac care. Utilizing a dataset comprised of various clinical parameters, our objective is to develop a predictive model that can effectively identify individuals at risk of heart failure. The integration of computational data science into healthcare promises not only to revolutionize diagnostic procedures but also to pave the way for more personalized and timely medical interventions. Through this project, we endeavor to contribute to the advancement of medical technologies and the betterment of patient care outcomes.

Dataset Description:

Our dataset, sourced from Kaggle, is curated to predict the risk of heart disease in a patient. It comprises various attributes related to patients' health and lifestyle. Understanding and analyzing this dataset are crucial for developing a predictive model to identify individuals at risk of heart disease. The dataset contains information on 12 attributes for each patient, with a total of 918 rows. We used *pandas* functionalities to study features' data types using 'DataFrame.info' and generated descriptive statistics using 'DataFrame.describe()'. This further helped us to understand the structure of our data. The dataset's target variable is 'HeartDisease', a binary attribute indicating the presence (1) or absence (0) of heart disease in the patient. *Table 1* provides a detailed description of each variable.

Data Cleaning

Using *pandas* functions such as 'isna()', 'isnull()', and 'DataFrame.duplicated()' to check for incompatible values, we discovered our dataset did not have such values, which was a significant advantage as it eliminated the need for a laborious data cleaning step.

Table 1: Heart Disease Dataset

#	Variable	Type	Description

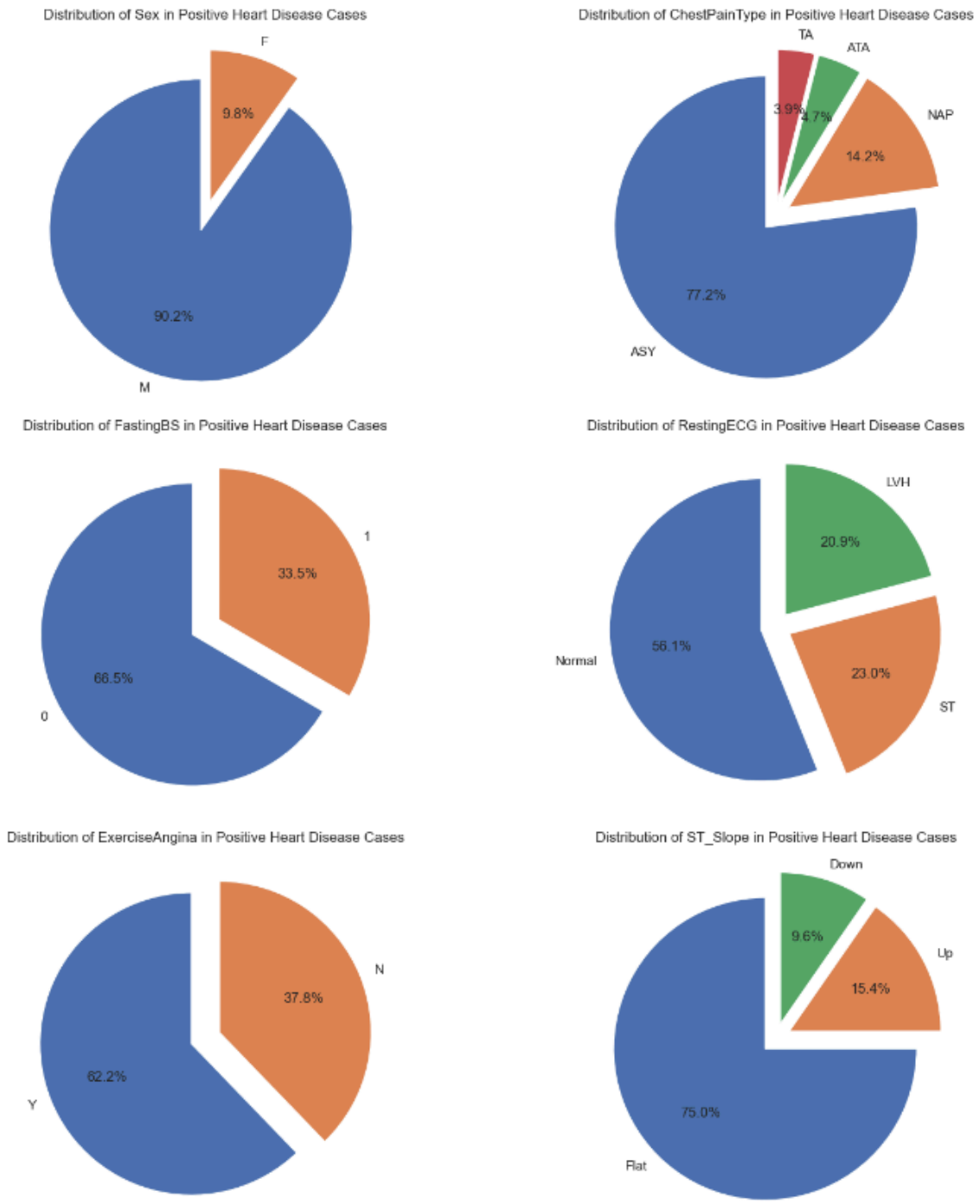
1.	Age	Numeric (Integer)	Age of the patient.
2.	Sex	Categorical ● M: Male ● F: Female	Gender of the patient
3.	ChestPainType	Categorical ● TA: Typical Angina ● ATA: Atypical Angina ● NAP: Non-Anginal Pain ● ASY: Asymptomatic	Type of chest pain
4.	RestingBP	Numeric (Integer)	Resting blood pressure of the patient (mm Hg)
5.	Cholesterol	Numeric (Integer)	Serum cholesterol level of the patient (mm/dl).
6.	FastingBS	Binary ● 1: if FastingBS > 120 mg/dl ● 0: otherwise	Fasting blood sugar
7.	RestingECG	Categorical ● Normal ● ST: ST-T wave abnormality ● LVH: Left ventricular hypertrophy	Resting electrocardiogram results
8.	MaxHR	Numeric (Integer)	Maximum heart rate achieved during exercise
9.	ExerciseAngina	Binary ● Y: Yes ● N: No	Exercise-induced angina
10.	Oldpeak	Numeric (Double)	ST depression measured during exercise
11.	ST_Slope	Categorical ● Up: upsloping ● Flat: flat ● Down: downsloping	Slope of the peak exercise ST segment
12.	HeartDisease	Binary ● 0: Heart Disease ● 1: No Heart Disease	Output class indicating the presence (1) or absence (0) of heart disease

.

Understanding the Data Set with Visualizations:

Categorical Variables:

Figure 1: Pie Charts for Categorical Variables in Positive Heart Disease Cases

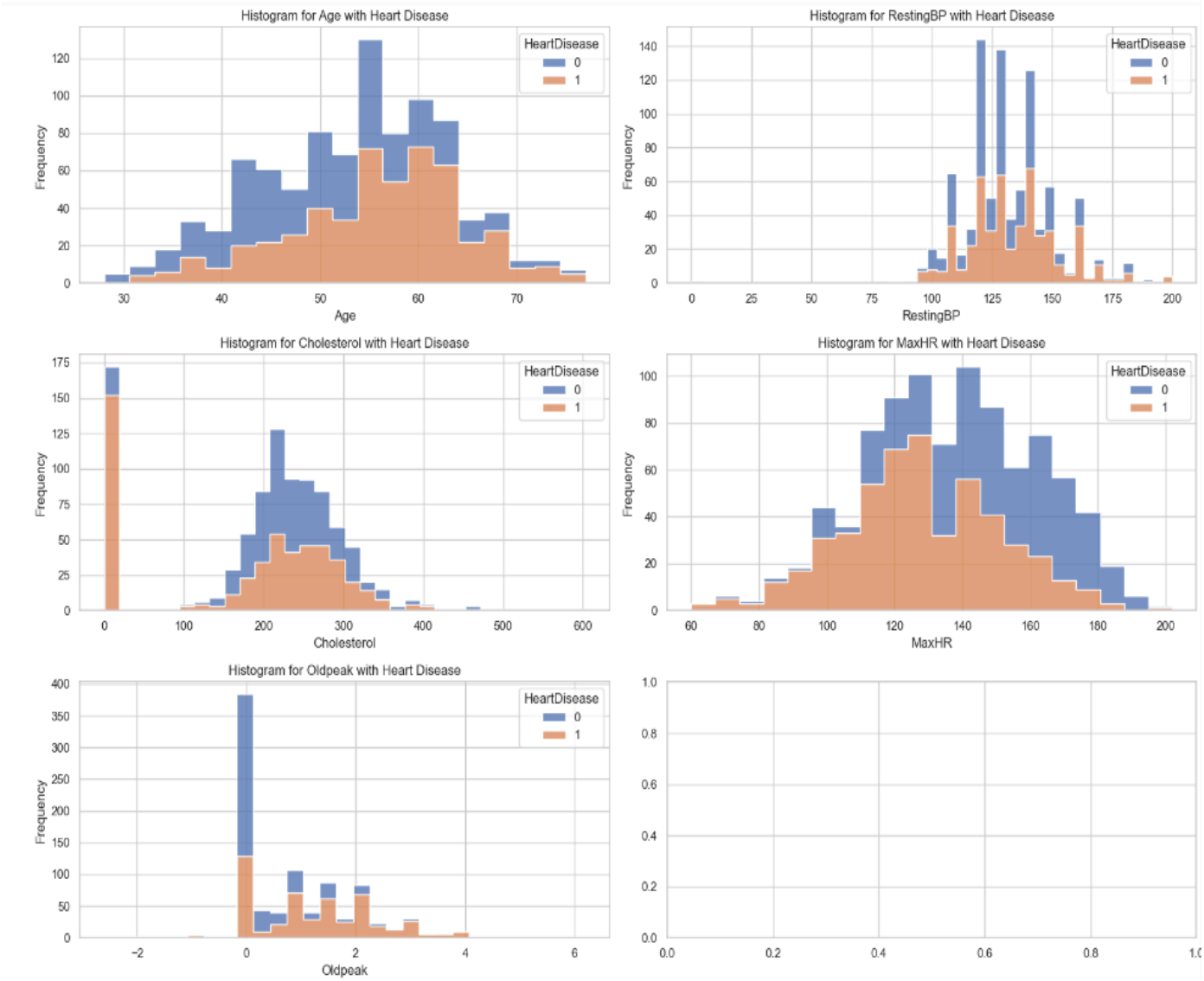


Observations from Figure 1:

- **Sex:** 90% of individuals diagnosed with heart disease are male, highlighting a significant gender disparity in the prevalence of this condition.
- **ChestPainType:** The ASY (asymptomatic) type of chest pain takes the lead, contributing to 77% of cases that eventually lead to heart diseases.
- **FastingBS:** Individuals with fasting blood sugar levels below 120 mg/dl demonstrate a heightened likelihood of developing heart diseases. This suggests that maintaining blood sugar levels within this range may be a crucial factor in preventing heart-related issues.
- **RestingECG:** 56% of individuals with a normal Resting ECG level exhibit a higher probability of heart diseases compared to those with LVH (Left Ventricular Hypertrophy) and ST (ST-T wave abnormality) levels.
- **ST_Slope:** A substantial 75% of cases with a flat ST_slope reading are associated with underlying heart problems.
- **ExerciseAngina:** For people with a heart disease, 37.8% are observed to have exercise induced angina, which is considerable amount.

Numerical Variables:

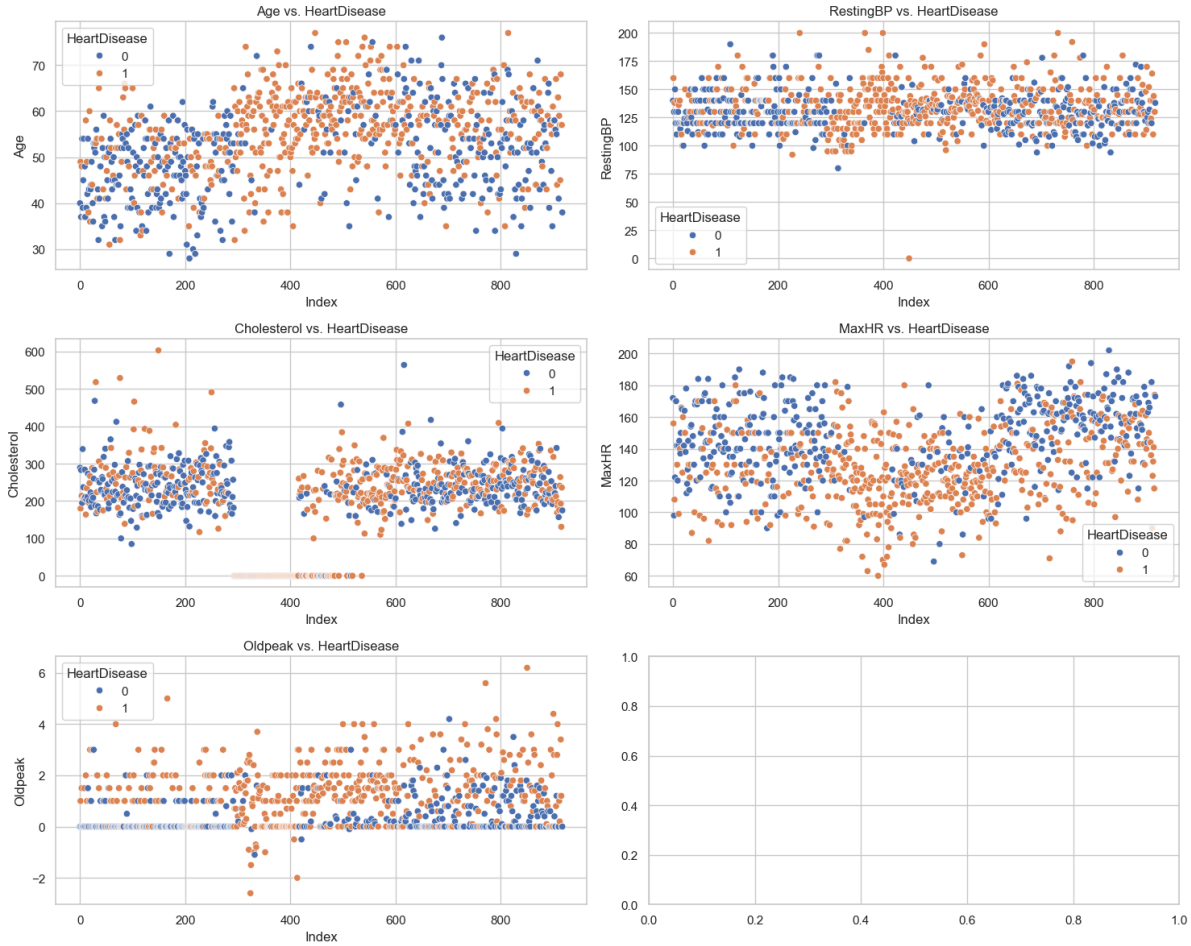
Figure 2: Histograms for Numerical Variables vs HeartDisease



Observations from Figure 2:

- **Age**: Appears to be fairly normally distributed with a slight right skew.
- **Resting Blood Pressure:** Shows a normal-like distribution with a few high values.
- **Cholesterol:** Shows normal-distribution with few outliers which needs to be considered. Many people have 0-50 mm/dl levels and then it drops and normal-distribution starts at 100mm/dl.
- **Maximum Heart Rate:** Fairly normal distribution with a left skew.
- **Oldpeak:** Mostly concentrated around lower values with a right skew.

Figure 3: Scatterplots for Numerical Variables vs HeartDisease



Observations from Figure 3:

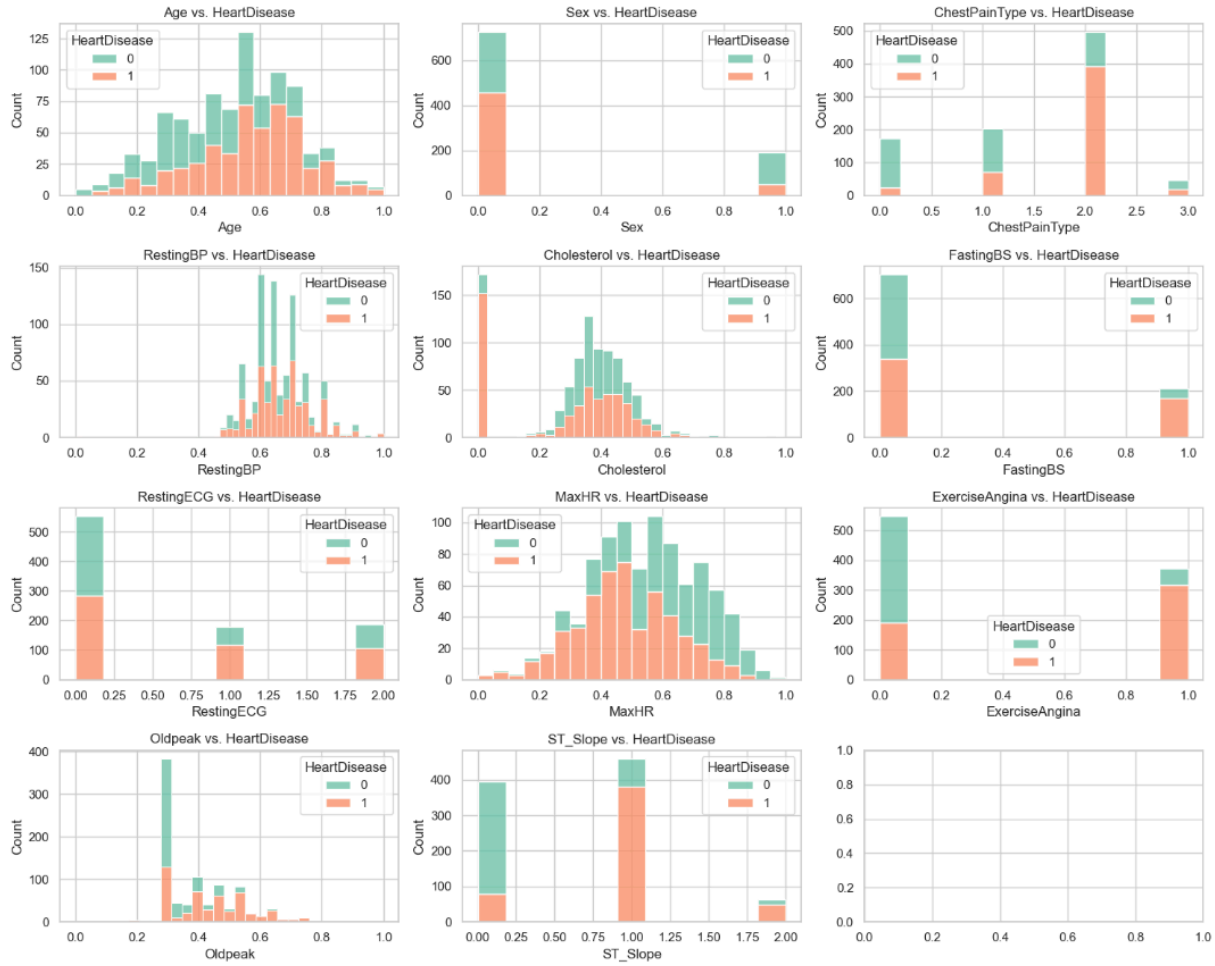
Age: It is observable that as age grows, so does the presence of a heart disease. This is evidenced by majority of orange points as a person gets older than 50 years.
RestingBP: The points are distributed across a wide range of blood pressure values, showing that resting blood pressure alone does not clearly segregate individuals with and without heart disease.
Cholesterol: Groups are scattered across the range of cholesterol levels, suggesting a complex relationship between cholesterol and heart disease where high cholesterol levels are present in individuals with and without heart disease. Additionally another very important observation is having a low cholesterol level close to 0 mm/dl, makes a heart disease certain.
MaxHR: There seems to be no clear separation between those with and without heart disease in terms of their maximum heart rate above a certain threshold(around 100 in *Figure 3*). But as observed, lower MaxHR evidences presence of a heart disease.
Oldpeak: The distribution of points shows some overlap, but there might be a slight trend where higher values of Oldpeak are more frequently associated with heart disease.

Features Engineering:

Scaling The Data:

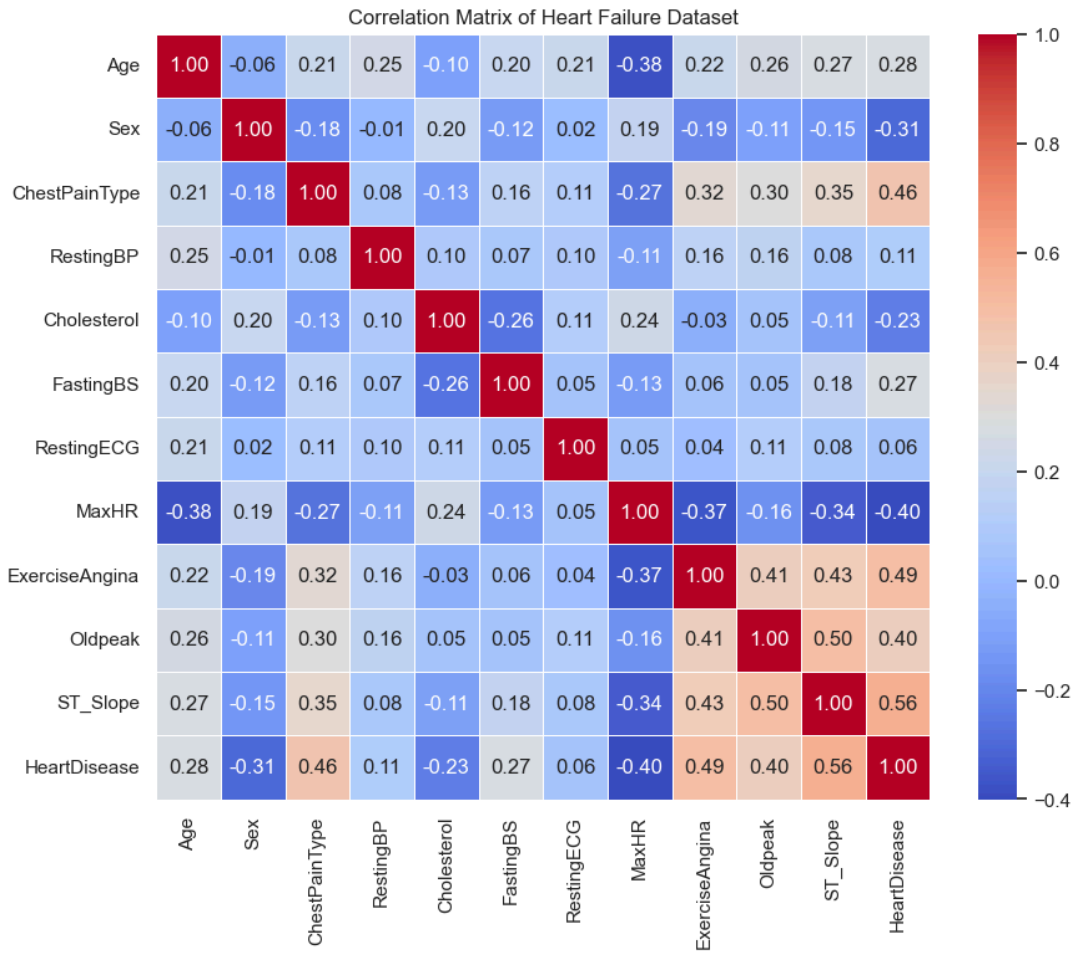
In the course of scaling the dataset, a crucial aspect involved the transformation of categorical variables into numerical formats, enhancing their compatibility with ML analytical processes. Notably, the 'Sex' variable, signifying gender, underwent a binary conversion—'M' (male) was assigned the numerical value 0, while 'F' (female) was designated as 1. Additionally, the 'ChestPainType' variable, representing diverse chest pain types, was systematically mapped to numerical values. Categories such as 'ATA' (Atypical Angina), 'NAP' (Non-Anginal Pain), 'ASY' (Asymptomatic), and 'TA' (Typical Angina) were respectively assigned numerical values 0, 1, 2, and 3. Similarly, the 'RestingECG' variable, indicating electrocardiographic results at rest, and the 'ST_Slope' variable, characterizing the slope of the ST segment during exercise, underwent categorical-to-numerical mappings. Meanwhile, the binary categorical variable 'ExerciseAngina' saw a straightforward conversion—'N' (No) was replaced with 0, and 'Y' (Yes) with 1. These mappings yield a structured numerical representation of categorical variables, laying the groundwork for effective use in later analyses and machine learning models.

Figure 4: Histograms for Normalized Categorial Features and Heart Disease vs Counts



Correlation Analysis:

Figure 5: Correlation Heatmap for all Normalized Features



Observations from Figure 5:

- The heatmap displays the correlation coefficients between different features. Values close to 1 or -1 indicate a strong positive or negative correlation, respectively.
- Notably, some features like Oldpeak, ExerciseAngina_Y, ChestPainType and ST_Slope_Flat show a more substantial correlation with HeartDisease, indicating their potential significance in predicting heart disease.

Feature Significance with SelectKBest():

Attributes	Score
ST_Slope	165.287287
ExerciseAngina	133.640134
ChestPainType	96.548821
Sex	67.640033
FastingBS	50.296983
MaxHR	8.738325
Oldpeak	5.546577
Age	5.191929
Cholesterol	4.957809
RestingECG	3.674137

Figure 6: Features Ranked by Chi-Squared Scores for Predicting Heart Disease

In order to enhance the process of feature engineering, we conducted an analysis to identify the features most relevant to predicting our target attribute, which in this case is Heart Disease. To accomplish this, we employed the 'SelectKBest()' class from the feature_selection library in scikit-learn. This class enables us to select the top k features based on their statistical significance. Specifically, we used the chi-squared statistical test to calculate scores for each feature. The chi-squared test helps assess the independence between categorical variables, providing valuable insights into the relationship between individual features and the occurrence of Heart Disease. The resulting scores (see *Figure 6*) offer a quantitative measure of the significance of each feature in our predictive model.

Dropping Features:

Based On Correlation and Feature Significance:

Analyzing and studying our correlation matrix (see *Figure 5*), HeartDisease vs normalized feature plots (*see Figure 4*), and SelectKbest table (see *Figure 6*) after putting them together, we made decisions to drop 'RestingECG' from our features list due to its low correlation with our target variable.

Despite of having a low score in the SelectKBest() table (see *Figure 6*), 'Cholesterol' exhibits noteworthy patterns in the Cholesterol vs. Heart Disease plot (*Figure 4*) and the Cholesterol vs. Frequency plot (*Figure 2*). Specifically, lower cholesterol levels, approaching 0 mm/dl, demonstrate a significant impact on the presence of Heart Disease. Similarly, despite of 'Age' having a low score in the SelectKBest() table*, Figures 2 and 4* indicate a discernible increase in the prevalence of Heart Disease with advancing Age. Finally, contrary to its low score in *Figure 6*, OldPeak exhibits a high correlation in *Figure 5*. Therefore, decided to keep these three features.

Based On Imbalance:

```
Population of Males and Females:
Sex
M    725
F    193
Name: count, dtype: int64
```

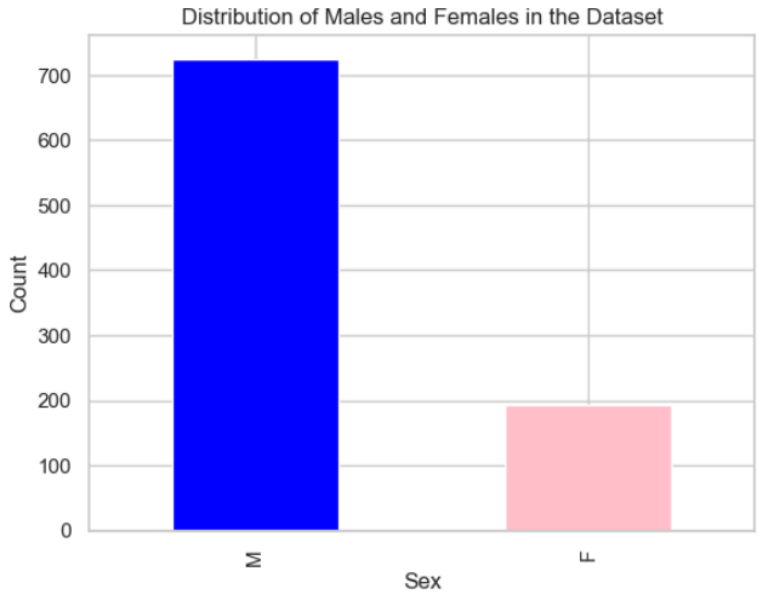


Figure 7

It is evident that we have a disproportionate number of subjects for each sex (see *Figure 7*), we decided to drop ‘Sex’ from our feature list as well.

Implementation and Evaluation of Machine Learning Models:

Preparing the training and validation data:

We prepared the data for our machine learning models by converting the categorial variables into equivalent numerical values as mentioned above. Additionally, we standardized the features using MinMaxScaler from scikit-learn, which normalizes the data, ensuring that all features contribute equally to the model's performance. Then we split the dataset into training (70%) and testing (30%) sets that helped us check our models’ accuracy score. The models used for the prediction of target are as follows:

Gradient Boosting Classifier:

The Gradient Boosting algorithm introduces an innovative method of weighting training data. Initially, all training points are assigned equal weights. As the training progresses, the algorithm adjusts these weights: increasing them for data points that are misclassified and decreasing them for correctly predicted points. This iterative process ensures that subsequent trees in the ensemble increasingly focus on challenging cases that were previously misclassified, thereby

enhancing the overall predictive power of the model.

Parameters: We selected 50 estimators with a maximum depth of 2 and a minimum sample leaf of 0.1. These parameters were chosen to balance the model's complexity and its ability to generalize.

Classification Report:

```
Training Score: 0.8722741433021807
Validation Score: 0.8695652173913043
Accuracy of the Gradient Boosting Classifier model: 0.8695652173913043
Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.85      0.84       112
           1       0.90      0.88      0.89       164

    accuracy                           0.87       276
   macro avg       0.86      0.87      0.87       276
weighted avg       0.87      0.87      0.87       276
```

Precision:

The model demonstrates a precision of 90% for Class 1 (Heart Disease) and 83% for Class 0 (No Heart Disease). A higher precision in Class 1 indicates its superior ability to accurately predict instances of heart failure compared to instances of no heart failure.

Recall Rates:

The recall rate for Class 1 is 88%, while for Class 0, it is 85%. This suggests that the model is slightly better at correctly identifying actual cases of heart failure compared to instances of no heart failure.

Overall Performance Inclination:

Evaluating the model's overall performance, it leans slightly towards accurately predicting instances of heart failure (Class 1). This inclination aligns with the balanced consideration of precision and recall rates.

Decision Tree Classfier:

For a simpler and much intrepretable model, we employed the Decision Tree which operates on a straightforward principle: it makes decisions based on the values of the input features, leading to a prediction at each leaf node.

Parameters: To prevent the common pitfall of overfitting, we optimized the Decision Tree by setting a maximum depth of 4. This control ensures that the tree does not grow overly complex,

which could lead to it being too tailored to the training data, at the expense of its ability to generalize to new data.

Classification Report:

```
Training Score: 0.8660436137071651
Validation Score: 0.8586956521739131
Accuracy of the Decision Tree model: 0.8586956521739131
Classification Report:
              precision    recall  f1-score   support

           0       0.79      0.88      0.84       112
           1       0.91      0.84      0.88       164

    accuracy                           0.86       276
   macro avg       0.85      0.86      0.86       276
weighted avg       0.86      0.86      0.86       276
```

Precision:

The Decision Tree model exhibits a precision of 79% for Class 0 (No Heart Disease) and 91% for Class 1 (Heart Disease). This indicates that the model is 79% accurate in predicting instances of no heart disease and 91% accurate in predicting instances of heart disease. The higher precision for Class 1 suggests a superior ability to accurately identify cases of heart disease compared to cases of no heart disease.

Recall Rates:

For Class 0, the recall rate is 88%, implying that the model correctly identifies 88% of the actual instances of no heart disease. On the other hand, the recall rate for Class 1 is 84%, indicating a relatively lower ability to capture all instances of heart disease. This suggests that the model is slightly better at correctly identifying cases of no heart disease compared to cases of heart disease.

Overall Performance Inclination:

Assessing the model's overall performance, it exhibits a slight inclination towards accurately predicting instances of heart disease (Class 1). This inclination aligns with the balanced consideration of precision and recall rates.

MLP Classifier:

We also used a Multi-Layer Perceptron (MLP) Classifier to leverage the capabilities of neural networks in capturing intricate patterns within the data. The MLP is well-suited for capturing intricate and non-linear patterns within the data. Its multiple hidden layers and interconnected nodes allow it to learn complex relationships, making it particularly effective when the underlying patterns in the data are intricate and cannot be easily captured by simpler models. Neural

networks, including MLPs, excel at modeling non-linear relationships and capturing interactions between different features.

Parameters:

The model was configured with 50 hidden layers and trained for a maximum of 1000 iterations. This choice of parameters helped us prevent overfitting and ensured that the MLP adapts to the underlying patterns without becoming overly complex. The validation set was then used to evaluate the model's performance.

Classification Report:

```
Training Score: 0.8769470404984424
Validation Score: 0.8731884057971014
Accuracy of the MLP Classifier model: 0.8731884057971014
Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.83      0.84       112
           1       0.89      0.90      0.89       164

    accuracy                           0.87       276
   macro avg       0.87      0.87      0.87       276
weighted avg       0.87      0.87      0.87       276
```

Precision:

The MLP Classifier model demonstrates a precision of 85% for Class 0 (No Heart Disease) and 89% for Class 1 (Heart Disease). This indicates that the model is 85% accurate in predicting instances of no heart disease and 89% accurate in predicting instances of heart disease. The higher precision for Class 1 suggests a superior ability to accurately identify cases of heart disease compared to cases of no heart disease.

Recall Rates:

For Class 0, the recall rate is 83%, implying that the model correctly identifies 83% of the actual instances of no heart disease. On the other hand, the recall rate for Class 1 is 90%, indicating a higher ability to capture all instances of heart disease. This suggests that the model is slightly better at correctly identifying cases of heart disease compared to cases of no heart disease.

Overall Performance Inclination:

Evaluating the model's overall performance, it exhibits a slight inclination towards accurately predicting instances of heart disease (Class 1). This inclination aligns with the balanced consideration of precision and recall rates.

Random Forest Classifier

Additionally, we leveraged ensemble learning through the use of a Random Forest Classifier. This model operates on the principle of aggregating multiple decision trees to enhance predictive accuracy and robustness. The Random Forest Classifier creates an ensemble of decision trees, each trained on a subset of the data with different combinations of features. By combining the predictions of these trees, the model achieves improved generalization, reducing the risk of overfitting and enhancing performance on unseen data.

Parameters:

The model was configured with 150 estimators (trees) in the ensemble, a maximum depth of 15 for each tree, and a minimum of 5 samples required for a leaf node, the model strikes a balance between complexity and interpretability.

Classification Report:

```
Training Score: 0.9127725856697819
Validation Score: 0.8804347826086957
Accuracy of the Random Forest Classifier model: 0.8804347826086957
Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.86      0.85       112
           1       0.90      0.90      0.90       164

    accuracy                           0.88       276
   macro avg       0.88      0.88      0.88       276
weighted avg       0.88      0.88      0.88       276
```

Precision:

The Random Forest Classifier model showcases a precision of 85% for Class 0 (No Heart Disease) and 90% for Class 1 (Heart Disease). This signifies that the model is 85% accurate in predicting instances of no heart disease and 90% accurate in predicting instances of heart disease. The higher precision for Class 1 suggests a superior ability to accurately identify cases of heart disease compared to cases of no heart disease.

Recall Rates:

For Class 0, the recall rate is 86%, implying that the model correctly identifies 86% of the actual instances of no heart disease. On the other hand, the recall rate for Class 1 is 90%, indicating a balanced ability to capture instances of heart disease. This suggests that the model performs well in correctly identifying both cases of heart disease and no heart disease.

Overall Performance Inclination:

Assessing the model's overall performance, it leans towards balanced accuracy in predicting instances of both heart disease and no heart disease. This inclination aligns with the well-balanced consideration of precision and recall rates.

K-Nearest Neighbours Classifier

We also explored the K Neighbors Classifier, a versatile algorithm relying on the proximity of data points for classification. Unlike parametric models, the K Neighbors Classifier leverages the simplicity and intuition of proximity-based decision making. It classifies a data point based on the class most prevalent among its nearest neighbors, making it particularly suitable for scenarios where similar instances tend to share the same class.

Parameters:

To prevent overfitting, we focused on optimizing the number of neighbors. A value of 10 was chosen to ensure a reasonable compromise between bias and variance. This choice allows the model to generalize well to new data, capturing local patterns without becoming overly sensitive to noise in the training set.

Classification Report:

```
Training Score: 0.8582554517133957
Validation Score: 0.8260869565217391
Accuracy of the K Neighbors Classifier model: 0.8260869565217391
Classification Report:
              precision    recall  f1-score   support

           0       0.74      0.88      0.80       112
           1       0.90      0.79      0.84       164

    accuracy                           0.83       276
   macro avg       0.82      0.83      0.82       276
weighted avg       0.84      0.83      0.83       276
```

Precision:

The K Neighbors Classifier model demonstrates a precision of 74% for Class 0 (No Heart Disease) and 90% for Class 1 (Heart Disease). This indicates that the model is 74% accurate in predicting instances of no heart disease and 90% accurate in predicting instances of heart disease. The higher precision for Class 1 suggests a superior ability to accurately identify cases of heart disease compared to cases of no heart disease.

Recall Rates:

For Class 0, the recall rate is 88%, implying that the model correctly identifies 88% of the actual instances of no heart disease. On the other hand, the recall rate for Class 1 is 79%, indicating a

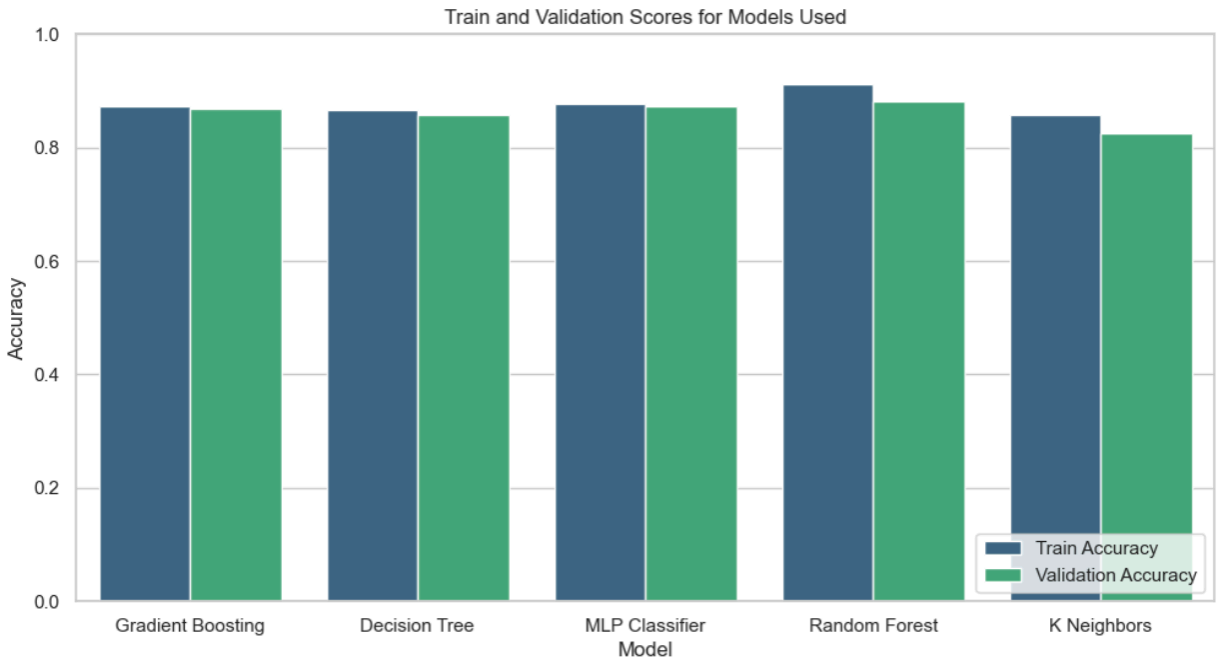
relatively lower ability to capture all instances of heart disease. This suggests that the model is slightly better at correctly identifying cases of no heart disease compared to cases of heart disease.

Overall Performance Inclination:

Evaluating the model's overall performance, it leans slightly towards accurately predicting instances of no heart disease (Class 0). This inclination aligns with the balanced consideration of precision and recall rates.

Visualizing the Training And Validation Scores Of Our ML Models

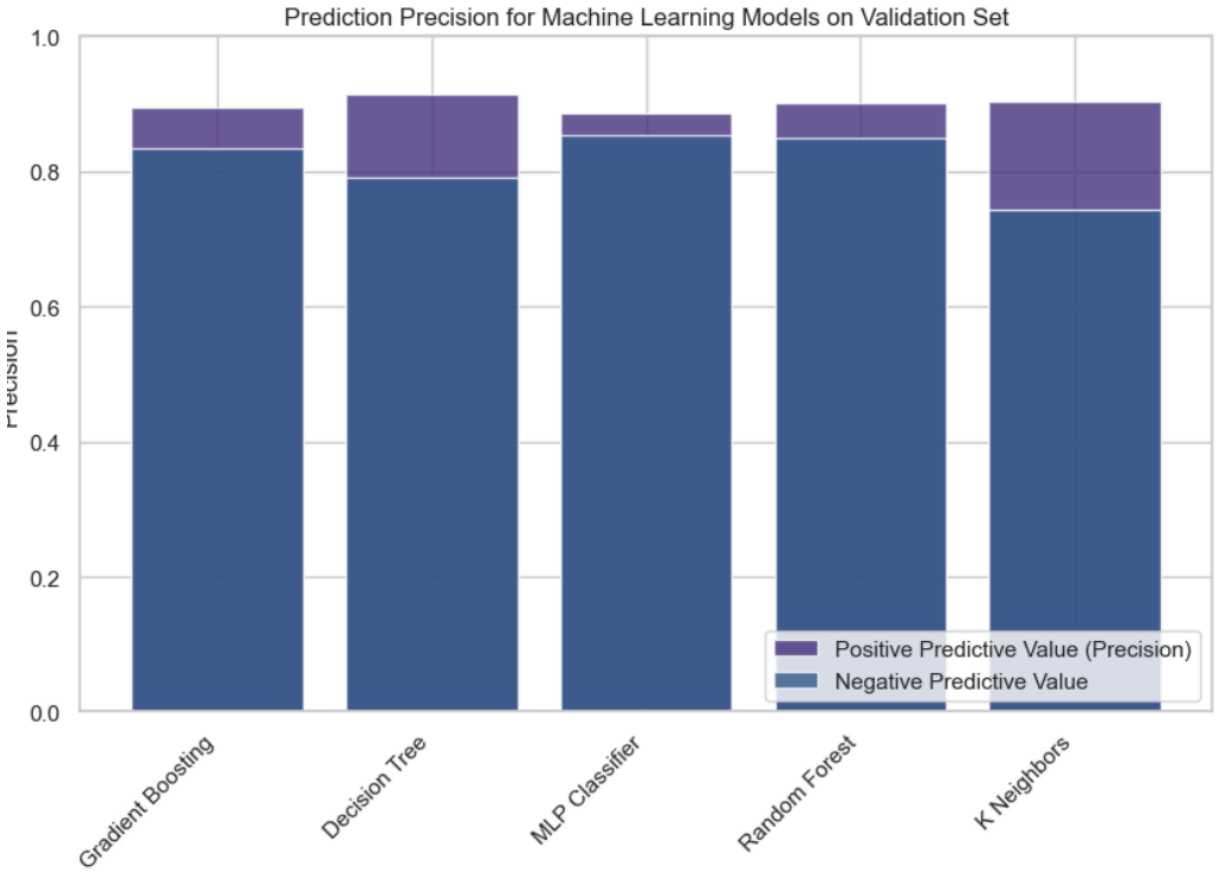
Figure 8: Comparing the Training and Validation Scores of the ML models



As seen in *Figure 8*, in all cases, the models exhibit a good balance between training and validation scores, suggesting that they are well-tuned and capable of making accurate predictions on new, unseen data. The close proximity between training and validation scores across all models is a positive indicator of robust model performance.

Analyzing The Predictive Performance Of Our ML Models

Figure 9: Comparing The Positive And Negative Predictive Performances



Positive Predictive Value (PPV), also known as Precision, is a measure that tells you the proportion of positive identifications made by a model that were actually correct. It is calculated as the number of true positives divided by the sum of true positives and false positives.

PPV = TP / (TP + FP)

Where:

- TP is the number of true positives (instances correctly identified as positive),
- FP is the number of false positives (instances incorrectly identified as positive).

Negative Predictive Value (NPV) is the proportion of negative identifications made by a model that were actually correct. It is calculated as the number of true negatives divided by the sum of true negatives and false negatives.

NPV = TN / (TN + FN)

Where:

- TN is the number of true negatives (instances correctly identified as negative),
- FN is the number of false negatives (instances incorrectly identified as negative).

From *Figure 9*, It appears that the machine learning models consistently exhibit higher Positive Predictive Values (PPV) for predicting heart failure (positive cases) compared to Negative Predictive Values (NPV) for predicting the absence of heart failure (negative cases). This implies that the models are more accurate in identifying individuals at risk of heart failure rather than accurately excluding those without heart failure.

Results Summary

Feature Importance:

Figure 3 illustrates that variables such as Age, MaxHR, Oldpeak, and Cholesterol significantly influence the likelihood of heart disease. Moreover, *Figure 5* underscores a substantial correlation between Oldpeak, ExerciseAngina_Y, ChestPainType, and ST_Slope_Flat with HeartDisease, suggesting their potential significance in predicting the condition. The observcations of features like ST_Slope, ExerciseAngina and ChestPainType show a considerably higher degree of heart disease prediction evidenced by all figures(4, 5 and 6).

Machine Learning Models Performance:

Gradient Boosting Classifier: Showed a slightly higher effectiveness in predicting heart failure, with a precision of 90% for heart disease and 83% for no heart disease, and recall rates of 88% and 85%, respectively.
Decision Tree Classifier: Displayed a slight inclination towards identifying no heart disease, with precision rates of 79% for no heart disease and 91% for heart disease.
MLP Classifier: Leaned slightly towards predicting heart disease, with a precision of 85% for no heart disease and 89% for heart disease, and recall rates of 83% and 90%, respectively.
Random Forest Classifier: Demonstrated balanced accuracy in predicting both conditions.
K-Nearest Neighbours Classifier: Was more accurate in predicting no heart disease.

Conclusion

Our analysis revealed significant insights into heart disease prediction. Key numerical variables like Age, MaxHR, and Oldpeak, along with categorical variables such as ExerciseAngina, ChestPainType, and ST_Slope, emerged as critical in understanding and predicting heart disease risk. The Gradient Boosting Classifier, with its high precision and recall, stood out among the models, showcasing the potential of machine learning in enhancing heart disease diagnostics.

The findings from both numerical and categorical analyses underscore the complexity of heart disease prediction and the need for a comprehensive approach that considers various physiological and symptomatic factors.

The visual and statistical analyses of the data revealed significant patterns and correlations, enhancing our understanding of the factors influencing heart disease. As a result, our project demonstrates the potential of machine learning in healthcare, particularly in early detection and preventive strategies for heart disease.

Future Work

While our current project successfully addressed the imbalance in male and female participants within the dataset, it is worthwhile to consider the integration of machine learning (ML) techniques for further refinement in future studies. Our approach, though effective, did not specifically leverage ML algorithms for balancing. In subsequent work, exploring methods such as oversampling, undersampling, synthetic data generation, and other ML-driven strategies could offer additional insights and improvements.